

Báo cáo bài tập thực hành tuần 2

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Lớp: 23TTH (Chiều thứ 6, ca 1)

Import các thư viện cần thiết

```
In [1]: # copy paste từ tuần 1 ==>))
import numpy as np
import pandas as pd
import scipy
from sklearn.preprocessing import StandardScaler, KBinsDiscretizer
from sklearn.decomposition import PCA
```

1. Khoảng cách giữa các điểm trong dữ liệu số

Trước hết, chúng ta sẽ lấy dữ liệu từ file `ionosphere.data` trực tiếp theo đường link dưới đây.

```
In [2]: link_hd_ko_che= r'https://archive.ics.uci.edu/ml/machine-learning-databases/ionosphere/ionosphere.data'
df = pd.read_csv(link_hd_ko_che, header=None)
df
```

```
Out[2]:
```

	0	1	2	3	4	5	6	7	8	9	...	25	26	...
0	1	0	0.99539	-0.05889	0.85243	0.02306	0.83398	-0.37708	1.00000	0.03760	...	-0.51171	0.41078	-0.461
1	1	0	1.00000	-0.18829	0.93035	-0.36156	-0.10868	-0.93597	1.00000	-0.04549	...	-0.26569	-0.20468	-0.184
2	1	0	1.00000	-0.03365	1.00000	0.00485	1.00000	-0.12062	0.88965	0.01198	...	-0.40220	0.58984	-0.221
3	1	0	1.00000	-0.45161	1.00000	1.00000	0.71216	-1.00000	0.00000	0.00000	...	0.90695	0.51613	1.000
4	1	0	1.00000	-0.02401	0.94140	0.06531	0.92106	-0.23255	0.77152	-0.16399	...	-0.65158	0.13290	-0.532
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
346	1	0	0.83508	0.08298	0.73739	-0.14706	0.84349	-0.05567	0.90441	-0.04622	...	-0.04202	0.83479	0.001
347	1	0	0.95113	0.00419	0.95183	-0.02723	0.93438	-0.01920	0.94590	0.01606	...	0.01361	0.93522	0.049
348	1	0	0.94701	-0.00034	0.93207	-0.03227	0.95177	-0.03431	0.95584	0.02446	...	0.03193	0.92489	0.025
349	1	0	0.90608	-0.01657	0.98122	-0.01989	0.95691	-0.03646	0.85746	0.00110	...	-0.02099	0.89147	-0.077
350	1	0	0.84710	0.13533	0.73638	-0.06151	0.87873	0.08260	0.88928	-0.09139	...	-0.15114	0.81147	-0.048

351 rows × 35 columns



Sau đó thực hiện lấy thông tin đặc trưng của bộ dữ liệu bằng hàm `describe()` và hàm `info()` của pandas.

```
In [3]: display(df.info())
display(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 351 entries, 0 to 350
Data columns (total 35 columns):
#   Column  Non-Null Count  Dtype
---  -
0   0        351 non-null    int64
1   1        351 non-null    int64
2   2        351 non-null    float64
3   3        351 non-null    float64
4   4        351 non-null    float64
5   5        351 non-null    float64
6   6        351 non-null    float64
7   7        351 non-null    float64
8   8        351 non-null    float64
9   9        351 non-null    float64
10  10       351 non-null    float64
11  11       351 non-null    float64
12  12       351 non-null    float64
13  13       351 non-null    float64
14  14       351 non-null    float64
15  15       351 non-null    float64
16  16       351 non-null    float64
17  17       351 non-null    float64
18  18       351 non-null    float64
19  19       351 non-null    float64
20  20       351 non-null    float64
21  21       351 non-null    float64
22  22       351 non-null    float64
23  23       351 non-null    float64
24  24       351 non-null    float64
25  25       351 non-null    float64
26  26       351 non-null    float64
27  27       351 non-null    float64
28  28       351 non-null    float64
29  29       351 non-null    float64
30  30       351 non-null    float64
31  31       351 non-null    float64
32  32       351 non-null    float64
33  33       351 non-null    float64
34  34       351 non-null    object
dtypes: float64(32), int64(2), object(1)
memory usage: 96.1+ KB
None
```

	0	1	2	3	4	5	6	7	8	
count	351.000000	351.0	351.000000	351.000000	351.000000	351.000000	351.000000	351.000000	351.000000	351.000000
mean	0.891738	0.0	0.641342	0.044372	0.601068	0.115889	0.550095	0.119360	0.511848	0.181110
std	0.311155	0.0	0.497708	0.441435	0.519862	0.460810	0.492654	0.520750	0.507066	0.483110
min	0.000000	0.0	-1.000000	-1.000000	-1.000000	-1.000000	-1.000000	-1.000000	-1.000000	-1.000000
25%	1.000000	0.0	0.472135	-0.064735	0.412660	-0.024795	0.211310	-0.054840	0.087110	-0.048110
50%	1.000000	0.0	0.871110	0.016310	0.809200	0.022800	0.728730	0.014710	0.684210	0.018110
75%	1.000000	0.0	1.000000	0.194185	1.000000	0.334655	0.969240	0.445675	0.953240	0.534110
max	1.000000	0.0	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000

8 rows × 34 columns



Bỏ cột cuối (vì yêu cầu trong file thực hành nó vậy)

```
In [4]: df.pop(df.columns[-1])
df
```

Out[4]:

	0	1	2	3	4	5	6	7	8	9	...	24	25	
0	1	0	0.99539	-0.05889	0.85243	0.02306	0.83398	-0.37708	1.00000	0.03760	...	0.56811	-0.51171	0.410
1	1	0	1.00000	-0.18829	0.93035	-0.36156	-0.10868	-0.93597	1.00000	-0.04549	...	-0.20332	-0.26569	-0.204
2	1	0	1.00000	-0.03365	1.00000	0.00485	1.00000	-0.12062	0.88965	0.01198	...	0.57528	-0.40220	0.589
3	1	0	1.00000	-0.45161	1.00000	1.00000	0.71216	-1.00000	0.00000	0.00000	...	1.00000	0.90695	0.516
4	1	0	1.00000	-0.02401	0.94140	0.06531	0.92106	-0.23255	0.77152	-0.16399	...	0.03286	-0.65158	0.132
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
346	1	0	0.83508	0.08298	0.73739	-0.14706	0.84349	-0.05567	0.90441	-0.04622	...	0.95378	-0.04202	0.834
347	1	0	0.95113	0.00419	0.95183	-0.02723	0.93438	-0.01920	0.94590	0.01606	...	0.94520	0.01361	0.935
348	1	0	0.94701	-0.00034	0.93207	-0.03227	0.95177	-0.03431	0.95584	0.02446	...	0.93988	0.03193	0.924
349	1	0	0.90608	-0.01657	0.98122	-0.01989	0.95691	-0.03646	0.85746	0.00110	...	0.91050	-0.02099	0.891
350	1	0	0.84710	0.13533	0.73638	-0.06151	0.87873	0.08260	0.88928	-0.09139	...	0.86467	-0.15114	0.811

351 rows × 34 columns

Khởi tạo các điểm point1, point2, point3 tương ứng là dòng 0, 1, 2 của array và tính chuẩn p = 1, 2, inf.

In [5]:

```
array = df.values
print(array)

point1 = array[0, :]
point2 = array[1, :]
point3 = array[2, :]

# p = 1
dist01_2 = np.linalg.norm(point1 - point2, 1)
dist01_3 = np.linalg.norm(point1 - point3, 1)

# p = 2
dist1_2 = np.linalg.norm(point1 - point2)
dist1_3 = np.linalg.norm(point1 - point3)

# p = inf
dist11_2 = np.linalg.norm(point1 - point2, np.inf)
dist11_3 = np.linalg.norm(point1 - point3, np.inf)

print(dist1_2)
print(dist1_3)
print(dist01_2)
print(dist01_3)
print(dist11_2)
print(dist11_3)
```

[[1. 0. 0.99539 ... -0.54487 0.18641 -0.453]
[1. 0. 1. ... -0.06288 -0.13738 -0.02447]
[1. 0. 1. ... -0.2418 0.56045 -0.38238]
...
[1. 0. 0.94701 ... 0.00442 0.92697 -0.00577]
[1. 0. 0.90608 ... -0.03757 0.87403 -0.16243]
[1. 0. 0.8471 ... -0.06678 0.85764 -0.06151]]
2.7763589251571923
1.1697276018372824
13.080950000000001
5.3597099999999999
1.12221
0.45772

Viết hàm tính các chuẩn p = 1, 2, inf cho 50 dòng đầu tiên của array trong mục 1

```
In [6]: def calculate_distances(array):
        distances = []
        for i in range(50):
            for j in range(i + 1, 50):
                point_i = array[i, :]
                point_j = array[j, :]
                dist_p1 = np.linalg.norm(point_i - point_j, 1)
                dist_p2 = np.linalg.norm(point_i - point_j, 2)
                dist_pinf = np.linalg.norm(point_i - point_j, np.inf)
                distances.append((i, j, dist_p1, dist_p2, dist_pinf))
        return distances

distances = calculate_distances(array)
for dist in distances:
    print(f"Distance between point {dist[0]} and point {dist[1]}: p=1: {dist[2]}, p=2: {dist[3]}, p=inf: {dist[4]}")
```

Distance between point 0 and point 1: p=1: 13.080950000000001, p=2: 2.7763589251571923, p=inf: 1.12221
Distance between point 0 and point 2: p=1: 5.359709999999999, p=2: 1.1697276018372824, p=inf: 0.45772
Distance between point 0 and point 3: p=1: 21.057290000000002, p=2: 4.772562679724594, p=inf: 1.6053600000000001
Distance between point 0 and point 4: p=1: 6.21387, p=2: 1.3773473382919792, p=inf: 0.53525
Distance between point 0 and point 5: p=1: 15.166310000000001, p=2: 3.049072399681582, p=inf: 0.97202
Distance between point 0 and point 6: p=1: 7.577039999999999, p=2: 1.530471119753653, p=inf: 0.48305
Distance between point 0 and point 7: p=1: 22.28699, p=2: 4.803640361215647, p=inf: 1.85243
Distance between point 0 and point 8: p=1: 5.81361, p=2: 1.2050178826473905, p=inf: 0.39934000000000003
Distance between point 0 and point 9: p=1: 17.20169, p=2: 3.5643139959184293, p=inf: 1.3090600000000001
Distance between point 0 and point 10: p=1: 7.06033, p=2: 1.5055474137668332, p=inf: 0.49585
Distance between point 0 and point 11: p=1: 16.25809, p=2: 3.584335524989255, p=inf: 1.47357
Distance between point 0 and point 12: p=1: 6.372240000000001, p=2: 1.3487913315261186, p=inf: 0.4608600000000005
Distance between point 0 and point 13: p=1: 14.074530000000001, p=2: 3.1139951292672245, p=inf: 1.42915
Distance between point 0 and point 14: p=1: 14.98318, p=2: 3.102336605818266, p=inf: 0.92012
Distance between point 0 and point 15: p=1: 18.11757, p=2: 3.851302322007453, p=inf: 1.6053600000000001
Distance between point 0 and point 16: p=1: 10.9936, p=2: 2.2606501964257983, p=inf: 0.70417
Distance between point 0 and point 17: p=1: 34.53319, p=2: 6.7281216338514565, p=inf: 2.0
Distance between point 0 and point 18: p=1: 12.67162, p=2: 2.739830121704628, p=inf: 0.9122399999999999
Distance between point 0 and point 19: p=1: 30.185820000000003, p=2: 6.0044294406213155, p=inf: 1.84356
Distance between point 0 and point 20: p=1: 6.24754, p=2: 1.4768651825403698, p=inf: 0.59114
Distance between point 0 and point 21: p=1: 30.92729, p=2: 6.163734316134984, p=inf: 2.0
Distance between point 0 and point 22: p=1: 7.7604299999999995, p=2: 1.624263827738585, p=inf: 0.54472
Distance between point 0 and point 23: p=1: 27.62719, p=2: 5.68832319052812, p=inf: 2.0
Distance between point 0 and point 24: p=1: 6.265499999999999, p=2: 1.386334658587168, p=inf: 0.5182800000000001
Distance between point 0 and point 25: p=1: 16.24697, p=2: 3.2079882195388434, p=inf: 1.0376
Distance between point 0 and point 26: p=1: 8.25909, p=2: 1.784492093145834, p=inf: 0.67808
Distance between point 0 and point 27: p=1: 25.77937, p=2: 5.358255846812468, p=inf: 1.99539
Distance between point 0 and point 28: p=1: 21.26855, p=2: 4.768385153959357, p=inf: 1.54487
Distance between point 0 and point 29: p=1: 27.856350000000003, p=2: 5.6610353045975605, p=inf: 2.0
Distance between point 0 and point 30: p=1: 8.44451, p=2: 1.9646133598497186, p=inf: 0.81359
Distance between point 0 and point 31: p=1: 20.84771, p=2: 4.389375054799943, p=inf: 1.54487
Distance between point 0 and point 32: p=1: 4.18114, p=2: 0.8691547915072435, p=inf: 0.29203999999999997
Distance between point 0 and point 33: p=1: 10.570170000000001, p=2: 2.636364973803893, p=inf: 1.4494500000000001
Distance between point 0 and point 34: p=1: 11.94792, p=2: 2.5758184746988673, p=inf: 1.07859
Distance between point 0 and point 35: p=1: 19.761910000000004, p=2: 4.020352814554961, p=inf: 1.36528
Distance between point 0 and point 36: p=1: 8.78373, p=2: 1.8330642644762896, p=inf: 0.6522600000000001
Distance between point 0 and point 37: p=1: 29.15677, p=2: 5.897289268799692, p=inf: 1.85243
Distance between point 0 and point 38: p=1: 10.47754, p=2: 2.172253563790379, p=inf: 0.6524099999999999
Distance between point 0 and point 39: p=1: 12.92207, p=2: 3.003011072024211, p=inf: 1.453
Distance between point 0 and point 40: p=1: 15.50779, p=2: 3.7555108229240934, p=inf: 1.27175
Distance between point 0 and point 41: p=1: 26.848529999999997, p=2: 5.654889629241228, p=inf: 1.99539
Distance between point 0 and point 42: p=1: 15.406070000000001, p=2: 3.4276336797125797, p=inf: 1.13039
Distance between point 0 and point 43: p=1: 19.812210000000004, p=2: 4.300753501573881, p=inf: 1.63825
Distance between point 0 and point 44: p=1: 6.320250000000001, p=2: 1.3083248824737685, p=inf: 0.4126300000000005
Distance between point 0 and point 45: p=1: 22.58643, p=2: 4.656229775247351, p=inf: 1.58212
Distance between point 0 and point 46: p=1: 6.06466, p=2: 1.4426177832676264, p=inf: 0.60167
Distance between point 0 and point 47: p=1: 15.984620000000001, p=2: 3.744958276563305, p=inf: 1.85243
Distance between point 0 and point 48: p=1: 7.74504, p=2: 1.590902529132442, p=inf: 0.5728000000000001
Distance between point 0 and point 49: p=1: 21.060719999999996, p=2: 4.400934784474772, p=inf: 1.6053600000000001
Distance between point 1 and point 2: p=1: 15.80038, p=2: 3.3800330447497107, p=inf: 1.10868
Distance between point 1 and point 3: p=1: 20.7696, p=2: 4.454509355899929, p=inf: 1.453
Distance between point 1 and point 4: p=1: 11.52616, p=2: 2.5405313555238793, p=inf: 1.0297399999999999
Distance between point 1 and point 5: p=1: 12.58796, p=2: 2.7838831970109665, p=inf: 1.02959
Distance between point 1 and point 6: p=1: 12.281089999999999, p=2: 2.7539302391854448, p=inf: 1.26313
Distance between point 1 and point 7: p=1: 22.639000000000003, p=2: 4.859195725220379, p=inf: 1.62237
Distance between point 1 and point 8: p=1: 16.54506, p=2: 3.4724946124940206, p=inf: 1.42077
Distance between point 1 and point 9: p=1: 14.54198, p=2: 3.2714733205392337, p=inf: 1.36117
Distance between point 1 and point 10: p=1: 16.563000000000002, p=2: 3.4588373100219676, p=inf: 1.4239600000000001
Distance between point 1 and point 11: p=1: 23.70878, p=2: 5.073432244723487, p=inf: 1.69707
Distance between point 1 and point 12: p=1: 17.698970000000003, p=2: 3.7733247031894828, p=inf: 1.45404
Distance between point 1 and point 13: p=1: 21.918319999999998, p=2: 4.540862974215364, p=inf: 1.51685
Distance between point 1 and point 14: p=1: 24.348249999999997, p=2: 4.871305851853279, p=inf: 1.51685
Distance between point 1 and point 15: p=1: 12.67514, p=2: 2.9013132908047004, p=inf: 1.18805
Distance between point 1 and point 16: p=1: 20.26597, p=2: 4.175570937309052, p=inf: 1.4589699999999999
Distance between point 1 and point 17: p=1: 36.99534, p=2: 7.017636574802089, p=inf: 2.0
Distance between point 1 and point 18: p=1: 19.97037, p=2: 3.9776427486263772, p=inf: 1.41224

Distance between point 1 and point 19: p=1: 30.94441, p=2: 5.969001458242408, p=inf: 2.0
Distance between point 1 and point 20: p=1: 17.83165000000003, p=2: 3.9084401599231375, p=inf: 1.51685
Distance between point 1 and point 21: p=1: 26.03694, p=2: 5.344618143141753, p=inf: 2.0
Distance between point 1 and point 22: p=1: 16.78962, p=2: 3.612240492215323, p=inf: 1.39219
Distance between point 1 and point 23: p=1: 29.96502, p=2: 6.035210277695384, p=inf: 2.0
Distance between point 1 and point 24: p=1: 17.14547, p=2: 3.7622455614034553, p=inf: 1.45077
Distance between point 1 and point 25: p=1: 22.28624, p=2: 4.658342934306146, p=inf: 1.51685
Distance between point 1 and point 26: p=1: 16.12232000000002, p=2: 3.494229059377762, p=inf: 1.39622
Distance between point 1 and point 27: p=1: 24.57698, p=2: 5.067302102302565, p=inf: 2.0
Distance between point 1 and point 28: p=1: 30.63362000000004, p=2: 6.0079292632819845, p=inf: 1.95099
Distance between point 1 and point 29: p=1: 31.388079999999995, p=2: 6.1176207054703875, p=inf: 2.0
Distance between point 1 and point 30: p=1: 20.0428, p=2: 4.331195102578502, p=inf: 1.51685
Distance between point 1 and point 31: p=1: 24.96278000000002, p=2: 5.202146473754849, p=inf: 1.62237
Distance between point 1 and point 32: p=1: 15.47241, p=2: 3.2536361298553342, p=inf: 1.338010000000001
Distance between point 1 and point 33: p=1: 18.6186, p=2: 4.146310474361513, p=inf: 1.69707
Distance between point 1 and point 34: p=1: 21.13621, p=2: 4.465606714378238, p=inf: 1.28522
Distance between point 1 and point 35: p=1: 25.97102, p=2: 5.401781547341581, p=inf: 1.78036
Distance between point 1 and point 36: p=1: 18.69414, p=2: 4.1204922206697585, p=inf: 1.48946
Distance between point 1 and point 37: p=1: 27.8386, p=2: 5.633787633200243, p=inf: 2.0
Distance between point 1 and point 38: p=1: 19.80253, p=2: 4.081736453802475, p=inf: 1.43733
Distance between point 1 and point 39: p=1: 21.84682, p=2: 4.3622179087248725, p=inf: 1.41003
Distance between point 1 and point 40: p=1: 22.70350000000002, p=2: 4.7064892360442085, p=inf: 1.62999
Distance between point 1 and point 41: p=1: 26.61692, p=2: 5.5594263374380635, p=inf: 2.0
Distance between point 1 and point 42: p=1: 20.28314, p=2: 4.178255657065518, p=inf: 1.43601
Distance between point 1 and point 43: p=1: 26.21146000000002, p=2: 5.367569491622815, p=inf: 1.64286
Distance between point 1 and point 44: p=1: 17.3333, p=2: 3.605786209330775, p=inf: 1.41497
Distance between point 1 and point 45: p=1: 19.992359999999998, p=2: 4.2578128887963125, p=inf: 1.67743
Distance between point 1 and point 46: p=1: 17.971229999999995, p=2: 3.889479524807914, p=inf: 1.46973
Distance between point 1 and point 47: p=1: 17.53049, p=2: 3.9049206328554233, p=inf: 1.93035
Distance between point 1 and point 48: p=1: 17.89603, p=2: 3.914337553750315, p=inf: 1.450540000000002
Distance between point 1 and point 49: p=1: 18.96035000000002, p=2: 3.966208782439472, p=inf: 1.63147
Distance between point 2 and point 3: p=1: 20.30478, p=2: 4.671267441562301, p=inf: 1.75535
Distance between point 2 and point 4: p=1: 8.50424, p=2: 1.8385075847545476, p=inf: 0.66143
Distance between point 2 and point 5: p=1: 14.24844000000002, p=2: 3.129187057815496, p=inf: 1.09924
Distance between point 2 and point 6: p=1: 11.57535, p=2: 2.3996959590123077, p=inf: 0.69877
Distance between point 2 and point 7: p=1: 21.16584, p=2: 4.67950618020748, p=inf: 1.75535
Distance between point 2 and point 8: p=1: 7.8223, p=2: 1.6954603740577365, p=inf: 0.5748
Distance between point 2 and point 9: p=1: 16.08502, p=2: 3.4523357405965025, p=inf: 1.20908
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Distance between point 34 and point 37: p=1: 28.33565, p=2: 6.181047497172304, p=inf: 2.0

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Distance between point 41 and point 49: p=1: 23.153529999999996, p=2: 5.07752101572214, p=inf: 1.8157299999999998
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Distance between point 44 and point 49: p=1: 23.96933, p=2: 5.083101493606044, p=inf: 1.89812
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Distance between point 45 and point 47: p=1: 23.68733, p=2: 5.360277165790963, p=inf: 2.0
Distance between point 45 and point 48: p=1: 25.00293, p=2: 5.428509774357968, p=inf: 1.88487
Distance between point 45 and point 49: p=1: 18.00217, p=2: 4.849467492797534, p=inf: 2.0
Distance between point 46 and point 47: p=1: 16.74438, p=2: 4.268348135286062, p=inf: 2.0
Distance between point 46 and point 48: p=1: 2.7493800000000004, p=2: 0.6808796879038175, p=inf: 0.26249999999999996
Distance between point 46 and point 49: p=1: 23.134939999999997, p=2: 5.038498819370706, p=inf: 1.95288
Distance between point 47 and point 48: p=1: 16.76954, p=2: 4.153002173223607, p=inf: 1.98103
Distance between point 47 and point 49: p=1: 19.93674, p=2: 4.810904677355393, p=inf: 2.0
Distance between point 48 and point 49: p=1: 21.72226, p=2: 4.8919122431826185, p=inf: 1.93369

Do yêu cầu đề bài tập là viết hàm tính các chuẩn $p = 1, 2, \text{inf}$ cho 50 dòng đầu tiên của array. Nên trong hàm 'calculate_distances' ta sẽ chạy vòng lặp for để tính khoảng cách giữa array với 50 dòng đầu tiên của array đó.

```
for i in range(50):
    for j in range(i + 1, 50):
        ... # Tính khoảng cách giữa dòng i và dòng j
    ...
```

Note: j bắt đầu từ $i + 1$ để tránh tính khoảng cách giữa một dòng với chính nó và tránh tính lại khoảng cách đã tính trước đó (do khoảng cách giữa dòng i và dòng j là như nhau với khoảng cách giữa dòng j và dòng i).

Tiếp theo chúng ta trích xuất dữ liệu của điểm i và j để tạo thành 2 vector `point_i` và `point_j` :

```
point_i = array[i]
point_j = array[j]
```

Sau đó, chúng ta sẽ tính khoảng cách giữa `point_i` và `point_j` theo các chuẩn $p = 1, 2, \text{inf}$... bằng cách sử dụng hàm `np.linalg.norm(vector, ord=p)` của NumPy: Với $p = 1$ là khoảng cách Manhattan, $p = 2$ là khoảng cách Euclidean, và $p = \text{inf}$ là khoảng cách Chebyshev. Cụ thể:

```
dist_p1 = np.linalg.norm(point_i - point_j, 1) # Khoảng cách Manhattan
dist_p2 = np.linalg.norm(point_i - point_j, 2) # Khoảng cách Euclidean
dist_pinf = np.linalg.norm(point_i - point_j, np.inf) # Khoảng cách Chebyshev
```

Sau khi tính toán xong thì ta lưu lại vào `distances` để sau này có thể sử dụng hoặc in ra kết quả.

```
distances.append((i, j, dist_p1, dist_p2, dist_pinf))
```

Cuối cùng ta được hàm `calculate_distances` như yêu cầu: input: array (dữ liệu đã được xử lý) output: distances (danh sách chứa khoảng cách giữa các điểm)

2. Độ đo tương đồng giữa các điểm trong tập dữ liệu phân loại

Trước hết, chúng ta sẽ lấy dữ liệu từ file `kddcup.data.corrected`

```
In [7]: df=pd.read_csv('D:\\kddcup.data.corrected',header=None)
df
```

```
Out[7]:
```

	0	1	2	3	4	5	6	7	8	9	...	32	33	34	35	36	37	38	39	40	41
0	0	tcp	http	SF	215	45076	0	0	0	0	...	0	0.0	0.0	0.00	0.00	0.0	0.00	0.0	0.0	normal.
1	0	tcp	http	SF	162	4528	0	0	0	0	...	1	1.0	0.0	1.00	0.00	0.0	0.00	0.0	0.0	normal.
2	0	tcp	http	SF	236	1228	0	0	0	0	...	2	1.0	0.0	0.50	0.00	0.0	0.00	0.0	0.0	normal.
3	0	tcp	http	SF	233	2032	0	0	0	0	...	3	1.0	0.0	0.33	0.00	0.0	0.00	0.0	0.0	normal.
4	0	tcp	http	SF	239	486	0	0	0	0	...	4	1.0	0.0	0.25	0.00	0.0	0.00	0.0	0.0	normal.
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
4898426	0	tcp	http	SF	212	2288	0	0	0	0	...	255	1.0	0.0	0.33	0.05	0.0	0.01	0.0	0.0	normal.
4898427	0	tcp	http	SF	219	236	0	0	0	0	...	255	1.0	0.0	0.25	0.05	0.0	0.01	0.0	0.0	normal.
4898428	0	tcp	http	SF	218	3610	0	0	0	0	...	255	1.0	0.0	0.20	0.05	0.0	0.01	0.0	0.0	normal.
4898429	0	tcp	http	SF	219	1234	0	0	0	0	...	255	1.0	0.0	0.17	0.05	0.0	0.01	0.0	0.0	normal.
4898430	0	tcp	http	SF	219	1098	0	0	0	0	...	255	1.0	0.0	0.14	0.05	0.0	0.01	0.0	0.0	normal.

4898431 rows × 42 columns

Lấy các cột có thuộc tính là `object`

```
In [8]: catagorical_columns = df.select_dtypes(include=['object']).columns
print("Catagorical columns:", catagorical_columns)
catagorical_data = df[catagorical_columns]
print("Catagorical data:", catagorical_data)
```

Catagorical columns: Index([1, 2, 3, 41], dtype='int64')

Catagorical data:

	1	2	3	41
0	tcp	http	SF	normal.
1	tcp	http	SF	normal.
2	tcp	http	SF	normal.
3	tcp	http	SF	normal.
4	tcp	http	SF	normal.
...	...	...	...	...
4898426	tcp	http	SF	normal.
4898427	tcp	http	SF	normal.
4898428	tcp	http	SF	normal.
4898429	tcp	http	SF	normal.
4898430	tcp	http	SF	normal.

[4898431 rows x 4 columns]

Check những giá trị giống nhau và loại bỏ những cột có giá trị giống nhau ở tất cả các dòng (nếu có)

```
In [9]: print(catagorical_data.duplicated().sum())
```



```
df = categorical_data.drop_duplicates()
df
```

4897822

```
Out[9]:
```

	1	2	3	41
0	tcp	http	SF	normal.
1288	tcp	http	S2	normal.
1484	tcp	http	S1	normal.
2062	tcp	smtp	SF	normal.
2067	udp	domain_u	SF	normal.
...	...	...	...	...
4574130	tcp	bgp	REJ	neptune.
4575586	tcp	Z39_50	REJ	neptune.
4578870	tcp	smtp	SF	neptune.
4579297	tcp	other	RSTO	neptune.
4810949	tcp	telnet	RSTO	buffer_overflow.

609 rows × 4 columns

Lấy 100 dòng đầu tiên theo yêu cầu đề bài

```
In [10]: df_100 = df.head(100)
df_100
```

```
Out[10]:
```

	1	2	3	41
0	tcp	http	SF	normal.
1288	tcp	http	S2	normal.
1484	tcp	http	S1	normal.
2062	tcp	smtp	SF	normal.
2067	udp	domain_u	SF	normal.
...	...	...	...	...
264244	tcp	ssh	SF	ipsweep.
264361	tcp	smtp	RSTO	normal.
264368	tcp	smtp	RSTO	ipsweep.
264396	tcp	domain	REJ	ipsweep.
264413	tcp	gopher	REJ	ipsweep.

100 rows × 4 columns

Tiếp theo chúng ta tạo ma trận $n \times n$ để lưu trữ độ đo tương đồng giữa các điểm, trong đó n là số dòng của dữ liệu (ở đây là 100 do đề bài).

- `overlap_matrix` để lưu trữ số lượng thuộc tính giống nhau giữa các điểm.
- `inv_freq_matrix` để lưu trữ độ đo tương đồng dựa trên tần suất nghịch đảo của các giá trị thuộc tính.

```
In [11]: n = len(df_100)
overlap_matrix = np.zeros((n, n))
inv_freq_matrix = np.zeros((n, n))
```

Tiếp theo ta tính độ tương đồng giữa các điểm trong tập dữ liệu với công thức tổng quát như sau:

$$\text{Sim}(\overline{X}, \overline{Y}) = \sum_{i=1}^d w_i \cdot S(x_i, y_i)$$

Với

- $w_i = \frac{1}{d}$ với $i = 1, 2, \dots, d$
- $S(x_i, y_i)$ là hàm đánh giá sự giống nhau giữa thuộc tính i được chia làm 2 loại là độ đo Overlap và độ đo dựa trên tần suất xuất hiện ngược

Ta tính độ đo Overlap với công thức:

$$\text{Overlap}(x_i, y_i) = S(x_i, y_i) = \begin{cases} 1 & \text{nếu } x_i = y_i \\ 0 & \text{nếu } x_i \neq y_i \end{cases}$$

Cụ thể với code như sau: `d = len(X)` để lấy số lượng thuộc tính của điểm X (và Y do $|X| = |Y|$).

Tiếp theo ta tính độ đo Overlap giữa X và Y bằng cách đếm số lượng thuộc tính giống nhau giữa hai điểm và chia cho tổng số thuộc tính d như công thức tổng quát sau khi thay $S(x_i, y_i)$ bằng công thức Overlap:

```
sum = 0
for i in range(d):
    if X[i] == Y[i]:
        sum += 1
return sum / d
```

```
In [12]: def overlap(X, Y):
        d = len(X)
        sum = 0
        for i in range(d):
            if X[i] == Y[i]:
                sum += 1
        return sum / d
```

Tương tự ta tính độ đo dựa trên tần suất xuất hiện ngược với công thức:

$$\text{Invfreq}(x_i, y_i) = S(x_i, y_i) = \begin{cases} \frac{1}{1+\log(f_i(x_i)) \cdot \log(f_i(y_i))} & \text{nếu } x_i = y_i \\ 0 & \text{nếu } x_i \neq y_i \end{cases}$$

Note: $\log$ ở đây là log cơ số 10, và $f_i(x_i)$ là tần suất xuất hiện của giá trị x_i trong thuộc tính i của tập dữ liệu

Tạo một hàm riêng để đếm và lấy tần suất xuất hiện của các giá trị thuộc tính trong tập dữ liệu để có thể sử dụng trong hàm tính độ đo dựa trên tần suất xuất hiện ngược. Bằng cách ta dùng kỹ thuật dictionary để lưu tên của thuộc tính và tần suất xuất hiện của các giá trị thuộc tính đó trong tập dữ liệu.

```
for i in range(d):
    unique, counts = np.unique(data[:, i], return_counts=True)
    freq_dict = {}
    for j in range(len(unique)):
        keyname = unique[j]
        value = counts[j]
        freq_dict[keyname] = value
    freqs.append(freq_dict)
return freqs
```

`unique` là mảng chứa các giá trị duy nhất của thuộc tính i , `counts` là mảng chứa số lần xuất hiện của từng giá trị đó. Ta tạo một dictionary `freq_dict` để lưu trữ tần suất xuất hiện của từng giá trị, với key là giá trị và value là số lần xuất hiện. Sau đó ta thêm `freq_dict` vào list `freqs` để có thể sử dụng sau này trong hàm tính độ đo dựa trên tần suất xuất hiện ngược.

```
In [13]: def freq(data):
        data = data.values
        freqs = []
        d = data.shape[1]

        for i in range(d):
            unique, counts = np.unique(data[:, i], return_counts=True)
```

```

freq_dict = {}
for j in range(len(unique)):
    keyname = unique[j]
    value = counts[j]
    freq_dict[keyname] = value
freqs.append(freq_dict)
return freqs

```

Cuối cùng ta áp dụng công thức tính độ đo dựa trên tần suất xuất hiện ngược vào hàm `inv_freq` bằng cách thay $S(x_i, y_i)$ bằng công thức đã cho và sử dụng tần suất xuất hiện từ `freqs`

```

In [14]: def inv_freq(X, Y, freqs):
    d = len(X)
    sim = 0

    for i in range(d):
        if X[i] == Y[i]:
            f_x = freqs[i].get(X[i], 0)
            f_y = freqs[i].get(Y[i], 0)

            if f_x > 0 and f_y > 0:
                sim += 1 / (1 + (np.log(f_x) * np.log(f_y)))
    return sim / d

```

```

In [15]: freqs = freq(df_100)

for i in range(n):
    for j in range(n):
        if i == j:
            overlap_matrix[i, j] = 1.0
            inv_freq_matrix[i, j] = 1.0
        else:
            overlap_matrix[i, j] = overlap(df_100.values[i], df_100.values[j])
            inv_freq_matrix[i, j] = inv_freq(df_100.values[i], df_100.values[j], freqs)

for i in range(n):
    overlap_scores = overlap_matrix[i].copy()
    inv_freq_scores = inv_freq_matrix[i].copy()

    overlap_scores[i] = -1
    inv_freq_scores[i] = -1

    best_overlap_idx = np.argmax(overlap_scores)
    best_inv_freq_idx = np.argmax(inv_freq_scores)

    print(f"\n df line {i}:")
    print(f" - Overlap ~= line {best_overlap_idx}, v = {overlap_scores[best_overlap_idx]}")
    print(f" - Inverse freq ~= line {best_inv_freq_idx}, v = {inv_freq_scores[best_inv_freq_idx]}")

```

```
df line 0:
- Overlap ~= line 1, v = 0.75
- Inverse freq ~= line 1, v = 0.07562948311289361

df line 1:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 0, v = 0.07562948311289361

df line 2:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 10, v = 0.11829495856778857

df line 3:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 18, v = 0.09210837115928414

df line 4:
- Overlap ~= line 12, v = 0.75
- Inverse freq ~= line 12, v = 0.15500833734913178

df line 5:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 0, v = 0.05338665208965101

df line 6:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 90, v = 0.1455904685833976

df line 7:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 10, v = 0.06571765147285336

df line 8:
- Overlap ~= line 13, v = 0.75
- Inverse freq ~= line 13, v = 0.11136202795077632

df line 9:
- Overlap ~= line 7, v = 0.75
- Inverse freq ~= line 7, v = 0.06529794218240816

df line 10:
- Overlap ~= line 2, v = 0.75
- Inverse freq ~= line 2, v = 0.11829495856778857

df line 11:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 48, v = 0.12493543449189934

df line 12:
- Overlap ~= line 4, v = 0.75
- Inverse freq ~= line 4, v = 0.15500833734913178

df line 13:
- Overlap ~= line 8, v = 0.75
- Inverse freq ~= line 23, v = 0.20356584444452291

df line 14:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 30, v = 0.1460101778738428

df line 15:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 0, v = 0.07562948311289361

df line 16:
- Overlap ~= line 31, v = 0.75
- Inverse freq ~= line 31, v = 0.201598848264613

df line 17:
- Overlap ~= line 8, v = 0.75
- Inverse freq ~= line 8, v = 0.11136202795077632

df line 18:
```

```
- Overlap ~= line 2, v = 0.75
- Inverse freq ~= line 2, v = 0.11829495856778857

df line 19:
- Overlap ~= line 7, v = 0.75
- Inverse freq ~= line 34, v = 0.07464217889855139

df line 20:
- Overlap ~= line 7, v = 0.75
- Inverse freq ~= line 7, v = 0.06529794218240816

df line 21:
- Overlap ~= line 7, v = 0.75
- Inverse freq ~= line 7, v = 0.06529794218240816

df line 22:
- Overlap ~= line 7, v = 0.5
- Inverse freq ~= line 24, v = 0.07103362777734068

df line 23:
- Overlap ~= line 13, v = 0.75
- Inverse freq ~= line 13, v = 0.20356584444452291

df line 24:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 67, v = 0.11393149289208154

df line 25:
- Overlap ~= line 7, v = 0.75
- Inverse freq ~= line 33, v = 0.13020624866054573

df line 26:
- Overlap ~= line 13, v = 0.75
- Inverse freq ~= line 13, v = 0.20356584444452291

df line 27:
- Overlap ~= line 4, v = 0.5
- Inverse freq ~= line 4, v = 0.13393359396718832

df line 28:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 0, v = 0.05338665208965101

df line 29:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 30, v = 0.09210837115928414

df line 30:
- Overlap ~= line 14, v = 0.75
- Inverse freq ~= line 14, v = 0.1460101778738428

df line 31:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 16, v = 0.201598848264613

df line 32:
- Overlap ~= line 2, v = 0.75
- Inverse freq ~= line 2, v = 0.11829495856778857

df line 33:
- Overlap ~= line 25, v = 0.75
- Inverse freq ~= line 25, v = 0.13020624866054573

df line 34:
- Overlap ~= line 19, v = 0.75
- Inverse freq ~= line 25, v = 0.13020624866054573

df line 35:
- Overlap ~= line 25, v = 0.75
- Inverse freq ~= line 25, v = 0.13020624866054573

df line 36:
- Overlap ~= line 37, v = 0.75
```

```
- Inverse freq ~= line 76, v = 0.0930384200100165

df line 37:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 36, v = 0.08836863533449638

df line 38:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 80, v = 0.0930384200100165

df line 39:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 38, v = 0.08836863533449638

df line 40:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 41, v = 0.19785911243982524

df line 41:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 40, v = 0.19785911243982524

df line 42:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 43, v = 0.19785911243982524

df line 43:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 42, v = 0.19785911243982524

df line 44:
- Overlap ~= line 47, v = 0.75
- Inverse freq ~= line 78, v = 0.1272194859934866

df line 45:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 46, v = 0.19785911243982524

df line 46:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 45, v = 0.19785911243982524

df line 47:
- Overlap ~= line 19, v = 0.75
- Inverse freq ~= line 19, v = 0.07464217889855139

df line 48:
- Overlap ~= line 44, v = 0.75
- Inverse freq ~= line 77, v = 0.15493470529954081

df line 49:
- Overlap ~= line 44, v = 0.75
- Inverse freq ~= line 96, v = 0.10103289858498217

df line 50:
- Overlap ~= line 44, v = 0.75
- Inverse freq ~= line 82, v = 0.15493470529954081

df line 51:
- Overlap ~= line 38, v = 0.75
- Inverse freq ~= line 61, v = 0.09862413265069958

df line 52:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 83, v = 0.14694022672457516

df line 53:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 52, v = 0.14227044204905503

df line 54:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 84, v = 0.14694022672457516
```

```
df line 55:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 54, v = 0.14227044204905503

df line 56:
- Overlap ~= line 44, v = 0.75
- Inverse freq ~= line 85, v = 0.12493543449189934

df line 57:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 86, v = 0.14694022672457516

df line 58:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 57, v = 0.14227044204905503

df line 59:
- Overlap ~= line 44, v = 0.75
- Inverse freq ~= line 87, v = 0.12493543449189934

df line 60:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 88, v = 0.14694022672457516

df line 61:
- Overlap ~= line 51, v = 0.75
- Inverse freq ~= line 60, v = 0.14227044204905503

df line 62:
- Overlap ~= line 51, v = 0.75
- Inverse freq ~= line 6, v = 0.12493543449189934

df line 63:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 89, v = 0.14694022672457516

df line 64:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 63, v = 0.14227044204905503

df line 65:
- Overlap ~= line 51, v = 0.75
- Inverse freq ~= line 92, v = 0.18052410488266954

df line 66:
- Overlap ~= line 38, v = 0.75
- Inverse freq ~= line 38, v = 0.08836863533449638

df line 67:
- Overlap ~= line 24, v = 0.75
- Inverse freq ~= line 24, v = 0.11393149289208154

df line 68:
- Overlap ~= line 51, v = 0.75
- Inverse freq ~= line 51, v = 0.09862413265069958

df line 69:
- Overlap ~= line 66, v = 0.75
- Inverse freq ~= line 66, v = 0.08836863533449638

df line 70:
- Overlap ~= line 66, v = 0.75
- Inverse freq ~= line 93, v = 0.18052410488266954

df line 71:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 72, v = 0.19785911243982524

df line 72:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 71, v = 0.19785911243982524
```

```
df line 73:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 74, v = 0.19785911243982524

df line 74:
- Overlap ~= line 37, v = 0.75
- Inverse freq ~= line 73, v = 0.19785911243982524

df line 75:
- Overlap ~= line 0, v = 0.75
- Inverse freq ~= line 95, v = 0.11787524927734337

df line 76:
- Overlap ~= line 36, v = 0.75
- Inverse freq ~= line 36, v = 0.0930384200100165

df line 77:
- Overlap ~= line 48, v = 0.75
- Inverse freq ~= line 48, v = 0.15493470529954081

df line 78:
- Overlap ~= line 44, v = 0.75
- Inverse freq ~= line 44, v = 0.1272194859934866

df line 79:
- Overlap ~= line 19, v = 0.75
- Inverse freq ~= line 19, v = 0.07464217889855139

df line 80:
- Overlap ~= line 38, v = 0.75
- Inverse freq ~= line 38, v = 0.0930384200100165

df line 81:
- Overlap ~= line 3, v = 0.75
- Inverse freq ~= line 97, v = 0.09411483193159714

df line 82:
- Overlap ~= line 50, v = 0.75
- Inverse freq ~= line 50, v = 0.15493470529954081

df line 83:
- Overlap ~= line 52, v = 0.75
- Inverse freq ~= line 52, v = 0.14694022672457516

df line 84:
- Overlap ~= line 54, v = 0.75
- Inverse freq ~= line 54, v = 0.14694022672457516

df line 85:
- Overlap ~= line 81, v = 0.75
- Inverse freq ~= line 98, v = 0.1480166386461558

df line 86:
- Overlap ~= line 57, v = 0.75
- Inverse freq ~= line 57, v = 0.14694022672457516

df line 87:
- Overlap ~= line 81, v = 0.75
- Inverse freq ~= line 99, v = 0.1480166386461558

df line 88:
- Overlap ~= line 60, v = 0.75
- Inverse freq ~= line 60, v = 0.14694022672457516

df line 89:
- Overlap ~= line 63, v = 0.75
- Inverse freq ~= line 63, v = 0.14694022672457516

df line 90:
- Overlap ~= line 6, v = 0.75
- Inverse freq ~= line 6, v = 0.1455904685833976

df line 91:
```

```
- Overlap ~= line 76, v = 0.75
- Inverse freq ~= line 76, v = 0.05674287100314158
```

```
df line 92:
- Overlap ~= line 76, v = 0.75
- Inverse freq ~= line 65, v = 0.18052410488266954
```

```
df line 93:
- Overlap ~= line 76, v = 0.75
- Inverse freq ~= line 70, v = 0.18052410488266954
```

```
df line 94:
- Overlap ~= line 81, v = 0.75
- Inverse freq ~= line 82, v = 0.1480166386461558
```

```
df line 95:
- Overlap ~= line 75, v = 0.75
- Inverse freq ~= line 78, v = 0.12030141934010158
```

```
df line 96:
- Overlap ~= line 3, v = 0.75
- Inverse freq ~= line 49, v = 0.10103289858498217
```

```
df line 97:
- Overlap ~= line 49, v = 0.75
- Inverse freq ~= line 49, v = 0.10103289858498217
```

```
df line 98:
- Overlap ~= line 76, v = 0.75
- Inverse freq ~= line 85, v = 0.1480166386461558
```

```
df line 99:
- Overlap ~= line 76, v = 0.75
- Inverse freq ~= line 87, v = 0.1480166386461558
```

Ta lấy tần số xuất hiện của các giá trị thuộc tính trong tập dữ liệu bằng cách gọi hàm `freq` với dữ liệu đã được lấy 100 dòng đầu tiên:

```
freqs = freq(df_100)
```

Tiếp theo ta tính ma trận độ đo tương đồng dựa của 2 phương pháp overlap và tần suất xuất hiện ngược tương ứng cho 2 ma trận `overlap_matrix` và `inv_freq_matrix` bằng cách sử dụng vòng lặp for để tính độ đo tương đồng giữa các điểm trong tập dữ liệu. Cụ thể:

```
for i in range(n):
    for j in range(n):
        if i == j:
            overlap_matrix[i, j] = 1.0
            inv_freq_matrix[i, j] = 1.0
        else:
            overlap_matrix[i, j] = overlap(df_100.values[i], df_100.values[j])
            inv_freq_matrix[i, j] = inv_freq(df_100.values[i], df_100.values[j], freqs)
```

Cuối cùng, ta có dùng 2 ma trận đó để tìm điểm tương đồng nhất với mỗi điểm trong tập dữ liệu bằng cách tìm chỉ số của giá trị lớn nhất trong mỗi hàng của ma trận như sau:

```
for i in range(n):
    overlap_scores = overlap_matrix[i].copy()
    inv_freq_scores = inv_freq_matrix[i].copy()

    overlap_scores[i] = -1
    inv_freq_scores[i] = -1

    best_overlap_idx = np.argmax(overlap_scores)
    best_inv_freq_idx = np.argmax(inv_freq_scores)

    print(f"\n df line {i}:")
    print(f"- Overlap ~= line {best_overlap_idx}, v = {overlap_scores[best_overlap_idx]}")
    print(f"- Inverse freq ~= line {best_inv_freq_idx}, v = {inv_freq_scores[best_inv_freq_idx]}")
```


Note: Ta gán giá trị -1 cho phần tử tại vị trí i trong `overlap_scores` và `inv_freq_scores` để đảm bảo rằng khi tìm chỉ số của giá trị lớn nhất, nó sẽ không chọn chính điểm đó mà sẽ chọn điểm khác có độ tương đồng cao nhất. Sau đó ta in ra kết quả với chỉ số của điểm tương đồng nhất và giá trị tương đồng tương ứng.